Supplement to: Fujii T, Salanti G, Belletti A, et al. Effect of adjunctive vitamin C, glucocorticoids, and vitamin B1 on longer term mortality in adults with sepsis or septic shock: a systematic review and a component network meta-analysis.

Supplement 3. eMethods

1. Assessment of risk of bias

The following instructions were provided to the judges.

Random sequence generation

Select '**Low**' if the investigators describe a random component in the sequence generation process such as:
- Referring to a random number table
- Using a computer random number generator;
- Coin tossing; Shuffling cards or envelopes; Throwing dice; or Drawing of lots
- Minimization, minimization may be implemented without a random element, and this is considered to be equivalent to being random.

Select '**High**' if the investigators describe a non-random component in the sequence generation process. Usually, the description would involve some systematic, non-random approach, for example
- Sequence generated by odd or even date of birth;
- Sequence generated by some rule based on date (or day) of admission;
- Sequence generated by some rule based on hospital or clinic record number

Other non-random approaches happen much less frequently than the systematic approaches mentioned above and tend to be obvious. They usually involve judgement or some method of non-random categorization of participants.

Select '**Unclear**' was selected if insufficient information about the sequence generation process to permit judgement of 'Low risk' or 'High risk'.

Allocation concealment

Select '**Low**' if participants and investigators enrolling participants could not foresee assignment because one of the following, or an equivalent method, was used to conceal allocation:
- Central allocation (including telephone, web-based and pharmacy-controlled randomization);
- Sequentially numbered drug containers of identical appearance;
- Sequentially numbered, opaque, sealed envelopes.

Select '**High**' if participants or investigators enrolling participants could possibly foresee assignments and thus introduce selection bias, such as allocation based on:
- Using an open random allocation schedule (e.g. a list of random numbers);
- Assignment envelopes were used without appropriate safeguards (e.g. if envelopes were unsealed or non-opaque or not sequentially numbered);
- Alternation or rotation;
- Date of birth; case record number; or any other explicitly unconcealed procedure.

Select '**Unclear**' if insufficient information to permit judgement of 'Low risk' or 'High risk'. This is usually the case if the method of concealment is not described or not described in sufficient detail to allow a definite judgement - for example if the use of assignment envelopes is described, but it remains unclear whether envelopes were sequentially numbered, opaque and sealed.

Blinding of study personnel and participants

Please assess the risk of bias for the all-cause mortality within 1 year but later than 90 days
Select '**Low**' if any one of the following:
- No blinding or incomplete blinding, but the review authors judge that the outcome, the all-cause mortality within 1 year but later than 90 days, is not likely to be influenced by lack of blinding;
- Blinding of participants and key study personnel ensured, and unlikely that the blinding could have been broken.

Select '**High**' if any one of the following:
- No blinding or incomplete blinding, and the outcome is likely to be influenced by lack of blinding;
- Blinding of key study participants and personnel attempted, but likely that the blinding could have been broken, and the outcome is likely to be influenced by lack of blinding.

Select '**Unclear**' if any one of the following:
- Insufficient information to permit judgement of 'Low risk' or 'High risk';
- The study did not address this outcome.

Blinding of outcome assessor of the all-cause mortality within 1 year but later than 90 days

Select '**Low**' if any one of the following:
- No blinding of outcome assessment, but the review authors judge that the outcome measurement is not likely to be influenced by lack of blinding, e.g. death;
- Blinding of outcome assessment ensured, and unlikely that the blinding could have been broken.

Select '**High**' if any one of the following:
- No blinding of outcome assessment, and the outcome measurement is likely to be influenced by lack of blinding;
- Blinding of outcome assessment, but likely that the blinding could have been broken, and the outcome measurement is likely to be influenced by lack of blinding.

Select '**Unclear**' if any one of the following:
- Insufficient information to permit judgement of 'Low risk' or 'High risk';
- The study did not address this outcome.

Incomplete outcome data on the all-cause mortality within 1 year but later than 90 days

Select '**Low**' if any one of the following:
- No missing outcome data;
- Reasons for missing outcome data unlikely to be related to true outcome (for survival data, censoring unlikely to be introducing bias);
- Missing outcome data balanced in numbers across intervention groups, with similar reasons for missing data across groups;
- The proportion of missing outcomes compared with observed event risk not enough to have a clinically relevant impact on the intervention effect estimate;
- Missing data have been imputed using appropriate methods.

Select '**High**' if any one of the following:

- Reason for missing outcome data likely to be related to true outcome, with either imbalance in numbers or reasons for missing data across intervention groups;
- The proportion of missing outcomes compared with observed event risk enough to induce clinically relevant bias in intervention effect estimate;
- 'As-treated' analysis done with substantial departure of the intervention received from that assigned at randomization;
- Potentially inappropriate application of simple imputation.

Select '**Unclear**' if any one of the following:
- Insufficient reporting of attrition/exclusions to permit judgement of 'Low risk' or 'High risk' (e.g. number randomized not stated, no reasons for missing data provided);
- The study did not address this outcome.

Selective reporting

Select '**Low**' if any of the following:
- The study protocol is available and all of the study's pre-specified (primary and secondary) outcomes that are of interest in the review have been reported in the pre-specified way;
- The study protocol is not available but it is clear that the published reports include all expected outcomes, including those that were pre-specified (convincing text of this nature may be uncommon).

Select '**High**' if any one of the following:
- Not all of the study's pre-specified primary outcomes have been reported;
- One or more primary outcomes is reported using measurements, analysis methods or subsets of the data (e.g. subscales) that were not pre-specified;
- One or more reported primary outcomes were not pre-specified (unless clear justification for their reporting is provided, such as an unexpected adverse effect);
- One or more outcomes of interest in the review are reported incompletely so that they cannot be entered in a meta-analysis;
- The study report fails to include results for a key outcome that would be expected to have been reported for such a study.

Select '**Unclear**' if insufficient information to permit judgement of 'Low risk' or 'High risk'. It is likely that the majority of studies will fall into this category.

Industrial sponsorship

Select '**Low**' if any of the following:
- The trial was funded by non-industrial funding
- The authors declared that there was no specific funding for the trial.

Select '**High**' if any of the following:
- The trial was funded by any party that is related to industries.
- Any one of the authors was supported by any party that was related to industries, unless the party was not committed to any aspect of the trial.

Select '**Unclear**' if any of the following
- The authors declared only 'no COI' and they did not report anything about the funding source.

- Insufficient information to permit the judgement of 'Low risk' or 'High risk'.

Fixed-size block randomisation in an unblinded study

Select '**Low**' if any of the following:
- the study was a blinded trial
- the randomisation was done in another way than block randomisation
- the study was an unblinded trial and various block sizes were used for the block randomisation.

Select '**High**' if the study was unblinded trial, and the block randomisation was done using a fixed size of a block.

Select '**Unclear**' if insufficient information to permit the judgement of 'Low risk' or 'High risk'.

Baseline imbalance

Select '**Low**' if any one of the following:
- Baseline was well-balanced;
- Baseline imbalance unlikely to be related to the outcome;

Select '**High**' if any one of the following:
- Baseline imbalance likely to be related to true outcome;

Select '**Unclear**' if any one of the following:
- Insufficient reporting of baseline imbalance to permit judgement of 'Low risk' or 'High risk'.

Inappropriate study deviation

Select '**Low**' if any one of the following:
- No deviation from the protocolised intervention;
- Reasons for protocol deviation unlikely to be related to true outcome;
- Protocol deviation balanced in numbers across intervention groups, with similar reasons;
- The proportion of protocol deviation compared with observed event risk not enough to have a clinically relevant impact on the intervention effect estimate;

Select '**High**' if any one of the following:
- Reason for protocol deviation likely to be related to true outcome, with either imbalance in numbers or reasons;
- The proportion of protocol deviation compared with observed event risk enough to induce clinically relevant bias in intervention effect estimate;

Select '**Unclear**' if any one of the following:
- Insufficient reporting of protocol compliance/deviation to permit judgement of 'Low risk' or 'High risk' (e.g. number randomized not stated, no reasons for missing data provided).

Fraud or concern that might be a source of risk of bias, if any

Report in free text if you have any concerns or suspicions on any aspect of the study.

2. Evaluation of the assumptions in the NMA

Transitivity assumption

A valid NMA satisfies the assumption of transitivity; that is, the available direct comparisons are similar in all respects other than the treatments being compared[1]. We evaluated the transitivity assumption by comparing the distribution of potential effect modifiers across treatment comparisons. The potential effect modifiers were predefined as follows: industrial sponsorship, blinding of the personnel, and vasopressor dependency of the study population.

We also estimated the prediction intervals in the results to express the impact of the common heterogeneity assumed across comparisons. We compared the estimated tau-squared value (the heterogeneity variance) with their empirical distributions[2, 3].

Consistency

The evaluation of transitivity was supplemented with the assessment of consistency with the statistical agreement between direct and indirect evidence. We used a side-splitting approach to evaluate inconsistency within each pairwise comparison as a local method and the design-by-treatment model as a global method to detect inconsistency in the network[4].

3. Changes from the protocol

A protocol for the study was published prior to the completion of the data collection[1]. The following changes were made to the protocol for the reasons specified.

- We had planned to close the database when the VITAMINS trial was published; however, we extended the study period to Jan 2021 as many trials being published so as to make the best of all available evidence.
- One of the sensitivity analyses was changed from “analysing only studies published in 2010 or after for all-cause mortality within one year but longer than 90 days” to “analysing only studies published in 2010 or after for all-cause mortality at the longest follow-up” due to the scarcity of available data.

4. References

1. Salanti G, Del Giovane C, Chaimani A, Caldwell DM, Higgins JP: **Evaluating the quality of evidence from a network meta-analysis**. *PLoS One* 2014, **9**(7):e99682.
2. Rhodes KM, Turner RM, Higgins JP: **Predictive distributions were developed for the extent of heterogeneity in meta-analyses of continuous outcome data**. *J Clin Epidemiol* 2015, **68**(1):52-60.
3. Turner RM, Davey J, Clarke MJ, Thompson SG, Higgins JP: **Predicting the extent of heterogeneity in meta-analysis, using empirical data from the Cochrane Database of Systematic Reviews**. *Int J Epidemiol* 2012, **41**(3):818-827.
4. Chaimani A, Caldwell D, Li T, Higgins J, Salanti G: **Chapter 11: Undertaking network meta-analyses.** In: *Cochrane Handbook for Systematic Reviews of Interventions.* version 6.2 (updated February 2021) edn. Edited by Higgins J, Thomas J, Chandler J, Cumpston M, Li T, Page M, Welch V: Cochrane; 2021.